

FPT UNIVERSITY - HO CHI MINH CITY CAMPUS

DEPARTMENT OF SEMICONDUCTOR ICs



FPT UNIVERSITY

CDP301 PROJECT REPORT

SEMICONDUCTOR INTEGRATED CIRCUIT DESIGN

**Project: “HANDHELD
DIGITAL OSCILLOSCOPE”**

GROUP 6

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Chapter 1

INTRODUCTION

1.1 PROBLEM STATEMENT AND SIGNIFICANCE

In the study and research of Semiconductor Integrated Circuit (IC) Design, observing, measuring, and analyzing electrical signals are fundamental requirements. However, commercial oscilloscopes available on the market are often bulky and prohibitively expensive, making it difficult for individual students to acquire them for personal practice or at-home circuit debugging.

Driven by these practical constraints, the project **”Handheld Digital Oscilloscope using ESP32”** is proposed to research and fabricate a compact, low-cost measurement device that still ensures the intuitive visualization of signal parameters such as amplitude (V_{pp}), period, and frequency in real time.

1.2 RESEARCH OBJECTIVES AND SCOPE

The system is designed to achieve the following core technical objectives:

- Reconstruct and display analog waveforms on an ILI9341 TFT screen via high-speed SPI communication (VSPI).
- Design an Analog Frontend (AFE) circuit incorporating an LM358 Op-Amp to generate a 1.65V static DC Bias, enabling the ESP32 microcontroller to acquire alternating current (AC) signals with negative amplitudes.

- Integrate an AC/DC coupling switch mechanism and a 1X/10X attenuation network to expand the safe measurement range for the hardware.

Chapter 2

THEORETICAL BACKGROUND AND TECHNOLOGY

To ensure accurate signal sampling and reconstruction, the hardware design is rooted in fundamental physical principles (First Principles) and differential mathematics.

2.1 IN-DEPTH COMPONENT ANALYSIS AND SELECTION RATIONALE

To ensure the oscilloscope meets the rigorous demands of real-time signal processing, the selection of core components and specific pin routing was meticulously analyzed. This section details the justification, environmental resilience, and digital extraction capabilities of the primary hardware.

2.1.1 ESP32 Dev Kit V1: The Processing and ADC Core

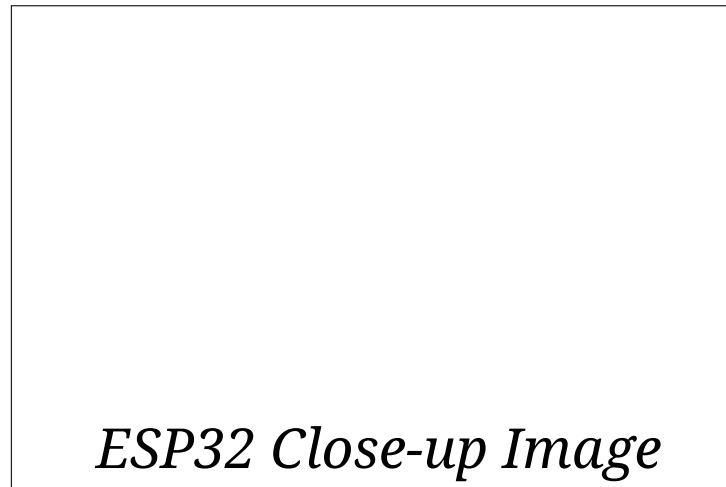


Figure 2.1: ESP32-WROOM-32 module featuring the EMI-shielded processing core.

The ESP32 was selected over traditional microcontrollers (such as the ATmega328P) due to its dual-core 240MHz processing density, which is mandatory for driving the SPI display without interrupting the ADC sampling routine.

Pin Selection and Analog-to-Digital Extraction: The analog signal is explicitly routed to pin D33. This is a critical design choice because D33 belongs to the **ADC1** channel of the ESP32. The ESP32's ADC2 channel shares its registers with the Wi-Fi module; thus, utilizing ADC2 causes critical sampling failures when wireless features are active. By utilizing ADC1, the system guarantees uninterrupted, deterministic data extraction. Furthermore, the 12-bit ADC provides a quantization density of 4096 levels—a 400% improvement in resolution compared to standard 10-bit (1024 levels) microcontrollers.

Environmental and Natural Resilience: The ESP32 module is enclosed in a metallic RF shield (Faraday cage). This structural feature significantly reduces electromagnetic interference (EMI) from external natural and artificial sources, preserving the integrity of the digital signals operating at high frequencies.

2.1.2 LM358 Operational Amplifier: The Analog Frontend

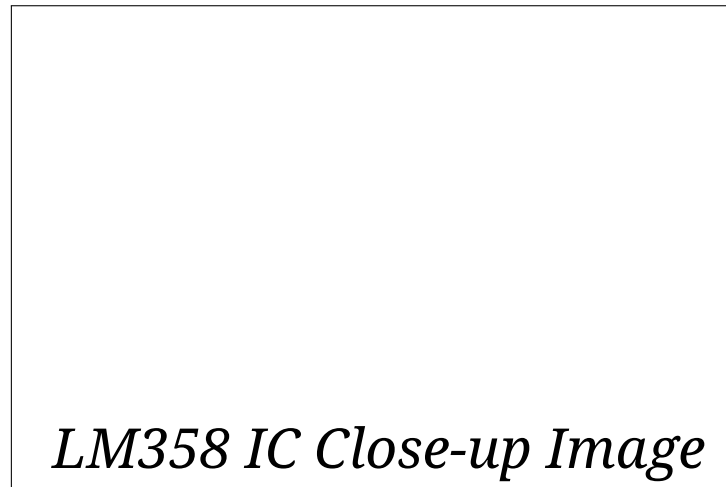


Figure 2.2: LM358 Dual Operational Amplifier integrated circuit.

The extraction of pure analog data requires pristine signal conditioning before it reaches the digital domain. The LM358 acts as the absolute barrier between the device under test (DUT) and the ESP32.

Role in Signal Extraction: The ESP32 ADC exhibits severe non-linearities near its absolute minimum ($0V$) and maximum ($3.3V$) thresholds. The LM358 creates a stable $1.65V$ virtual ground, strictly forcing the extracted AC signal to operate within the most linear region of the ESP32's ADC curve.

Resilience to External Anomalies: Analog signals are highly susceptible to thermal drift and common-mode noise (e.g., 50Hz hum from ambient AC power lines). The LM358 features a high Common-Mode Rejection Ratio (CMRR), which mathematically subtracts background environmental noise from the actual measured signal, ensuring that only the true differential voltage is digitized.

2.1.3 ILI9341 3.2-inch TFT: High-Density Data Visualization

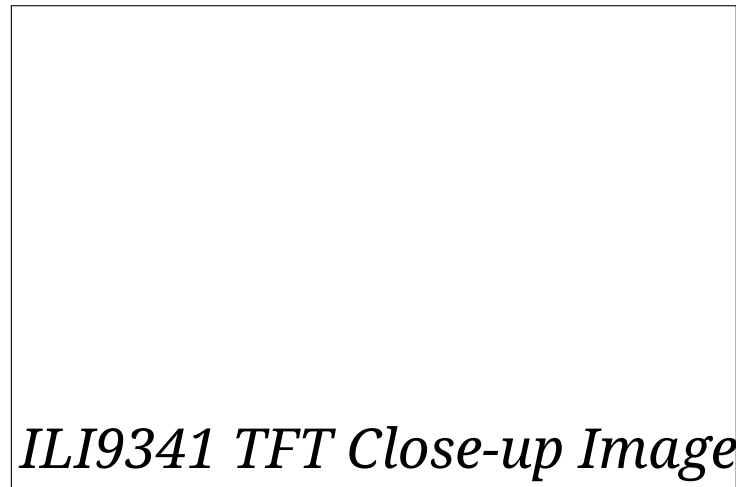


Figure 2.3: 3.2-inch ILI9341 TFT Display module.

Visualizing complex waveforms requires a high pixel density to prevent data aliasing on the screen. The ILI9341 provides a 320×240 resolution, offering a substantial physical plotting area.

Pin Routing and Interface Rationale: To update the screen at 30 Frames Per Second (FPS), the display is hardwired to the ESP32's Hardware VSPI pins (SCK: D18, MOSI: D23). Utilizing the hardware SPI bus, rather than software bit-banging, offloads the communication overhead from the CPU, allowing clock speeds up to 40MHz.

Environmental Consideration: The TFT module includes a dedicated 3.3V LDO regulator on its breakout board. Supplying it with a raw 5V rail ensures that the display backlight maintains constant brightness regardless of minor voltage drops caused by the ESP32's power consumption or battery degradation over time.

2.2 FIRST PRINCIPLES AND DIFFERENTIAL EQUATIONS

2.2.1 AC Coupling

Based on the capacitor current equation $i(t) = C \frac{dv_C(t)}{dt}$, the AC coupling stage acts as a high-pass filter defined by:

$$v_{in}(t) = RC \frac{dv_{out}(t)}{dt} + v_{out}(t) \quad (2.1)$$

2.2.2 Op-Amp Buffering

The LM358 maintains signal integrity via negative feedback, modeled by the differential equation:

$$\tau \frac{dv_{out}(t)}{dt} + v_{out}(t) = A_{OL}(v^+ - v^-) \quad (2.2)$$

This ensures an input impedance approaching infinity, protecting the integrity of the measured signal.

2.3 MATHEMATICAL FOUNDATION AND CIRCUIT THEORY

2.3.1 AC Coupling Principle via Capacitor

The AC Coupling mode operates based on the fundamental differential equation characterizing the voltage-current relationship of a capacitor:

$$i(t) = C \frac{dv_C(t)}{dt} \quad (2.3)$$

In the complex frequency domain (Laplace Domain), the impedance of the capacitor is expressed as $Z_C = \frac{1}{j\omega C}$. For the DC component ($\omega = 0$), the impedance approaches infinity, acting as a barrier that

blocks direct current. For the AC component ($\omega > 0$), the capacitor forms a first-order passive High-Pass Filter with the cutoff frequency f_c defined by:

$$f_c = \frac{1}{2\pi R_{eq}C} \quad (2.4)$$

In this project's practical implementation, a $1\mu F$ capacitor is placed in series on the signal path, combined with a bypass switch, to eliminate static noise from the external circuit when pure AC measurement is required.

2.3.2 Voltage Attenuation Network

To measure voltage levels exceeding the ESP32's 3.3V threshold, the system utilizes a 10X attenuation network (Voltage Divider). According to Ohm's law, the output voltage depends on the impedance ratio:

$$V_{out} = V_{in} \frac{R_1}{R_1 + R_2} \quad (2.5)$$

Based on actual components, the group utilizes a configuration where $R_2 = 100k\Omega$ is in series with $R_1 = 11k\Omega$ (a combination of $10k\Omega + 1k\Omega$). The voltage division ratio is approximately ≈ 0.099 , allowing the system to expand its measurement range by a factor of 10 with extremely low hardware error (under 1%).

2.3.3 Current Buffering and DC Bias with Op-Amp

The ESP32 microcontroller features a unipolar ADC capable of quantizing voltages strictly between $0V$ and $3.3V$. Any negative oscillation dropping below $0V$ will be clipped and poses a risk of damaging the IC pin. To resolve this issue, the LM358 Op-Amp is configured as a Voltage Follower combined with a voltage divider to create a Virtual Ground at:

$$V_{bias} = 3.3V \times \frac{10k\Omega}{10k\Omega + 10k\Omega} = 1.65V \quad (2.6)$$

The external AC signal, after passing through the filter capacitor, will "ride" on this static 1.65V DC level. Thanks to the nearly infinite input impedance ($Z_{in} \rightarrow \infty$) and near-zero output impedance ($Z_{out} \approx 0$) of the Op-Amp, the signal is stably buffered without causing voltage drops in the source circuit.

2.4 DIGITAL SIGNAL PROCESSING (DSP)

2.4.1 Nyquist-Shannon Sampling Theorem

To perfectly reconstruct a continuous waveform $x(t)$ on a digital display, the sampling process must strictly adhere to the Nyquist-Shannon theorem. The sampling frequency f_s of the ESP32's internal ADC must be at least twice the maximum frequency f_{max} of the signal being measured:

$$f_s \geq 2f_{max} \quad (2.7)$$

If this condition is violated, aliasing will occur, completely distorting the oscillation shape on the screen.

2.4.2 Quantization Error

The ESP32 is equipped with a 12-bit Analog-to-Digital Converter (ADC), equivalent to $2^{12} = 4096$ resolution levels. With a reference voltage $V_{ref} = 3.3V$, the error per Least Significant Bit (LSB) is:

$$Q = \frac{3.3V}{4096 - 1} \approx 0.805mV \quad (2.8)$$

This quantization error dictates the maximum physical accuracy limit the Oscilloscope system can achieve, accounting for the minute noise spikes always present on the micro-level graphs.

Chapter 3

SYSTEM DESIGN AND IMPLEMENTATION

3.1 OVERALL ARCHITECTURE BLOCK DIAGRAM

The system is divided into two isolated subsystems: the Analog Frontend (AFE) block for processing analog circuitry, and the Digital block responsible for numerical quantization and SPI communication.

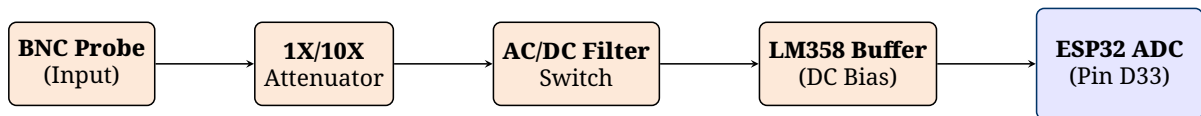


Figure 3.1: Signal chain architecture of the Handheld ESP32 Oscilloscope system.

3.2 SCHEMATIC DESIGN AND NOISE REDUCTION

3.2.1 Analog Frontend (AFE) Structure

At the raw signal input, a Zener Diode is connected in parallel to the ground to clip any transient overvoltage spikes before they reach the IN+ pin of the LM358 IC. A $10k\Omega$ resistor is placed in series at the IN- output of the LM358 before entering the ESP32's D33 ADC pin to limit current maximally, acting as a final safeguard for the

digital microcontroller core.

3.2.2 SPI Communication Design (TFT ILI9341)

To ensure real-time graph rendering speed (high FPS, no lag), the design utilizes the ESP32's default Hardware VSPI. The synchronization and data signals include the clock SCK (D18) and data MOSI (D23). The display's power pin VCC is supplied with a rated voltage of 5V instead of 3.3V, allowing the built-in **662K** LDO regulator on the back of the TFT to operate in its proper Drop-out mode, thereby providing clean power to the screen core.

3.2.3 Control Buttons and Input Pull-up

The push-button schematic is structured in an Active-LOW configuration, directly utilizing the ESP32's Internal Pull-up Resistors on pins D32 (Up), D25 (Down), D26 (OK), and D27 (Hold). This design completely eliminates external components, saving Printed Circuit Board (PCB) real estate to the maximum extent.

3.3 PCB LAYOUT AND 3D VISUALIZATION

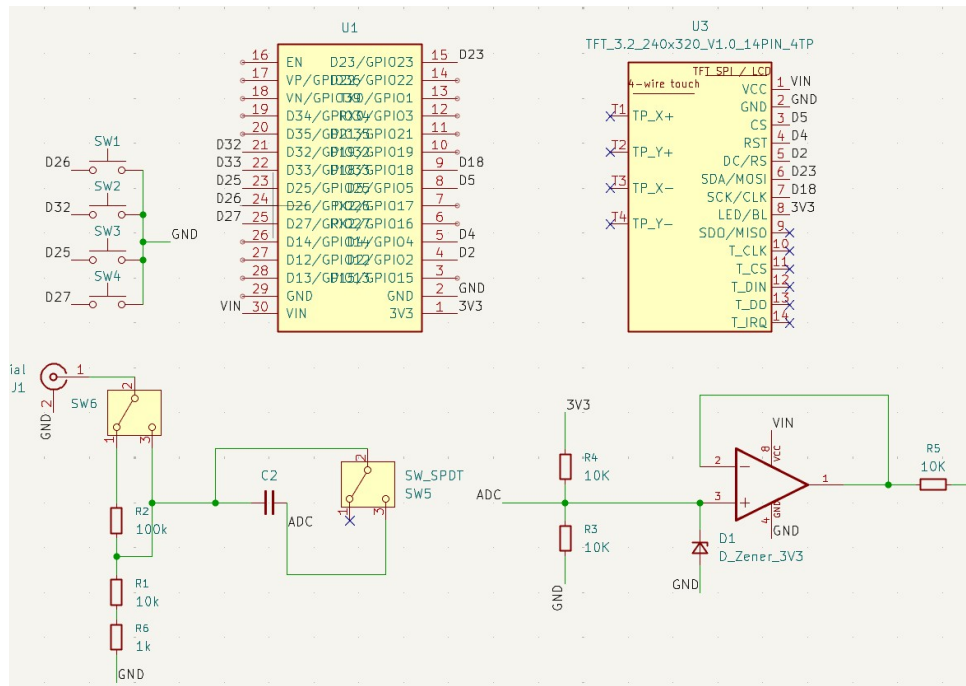


Figure 3.2: System schematic detailing the Analog Frontend.

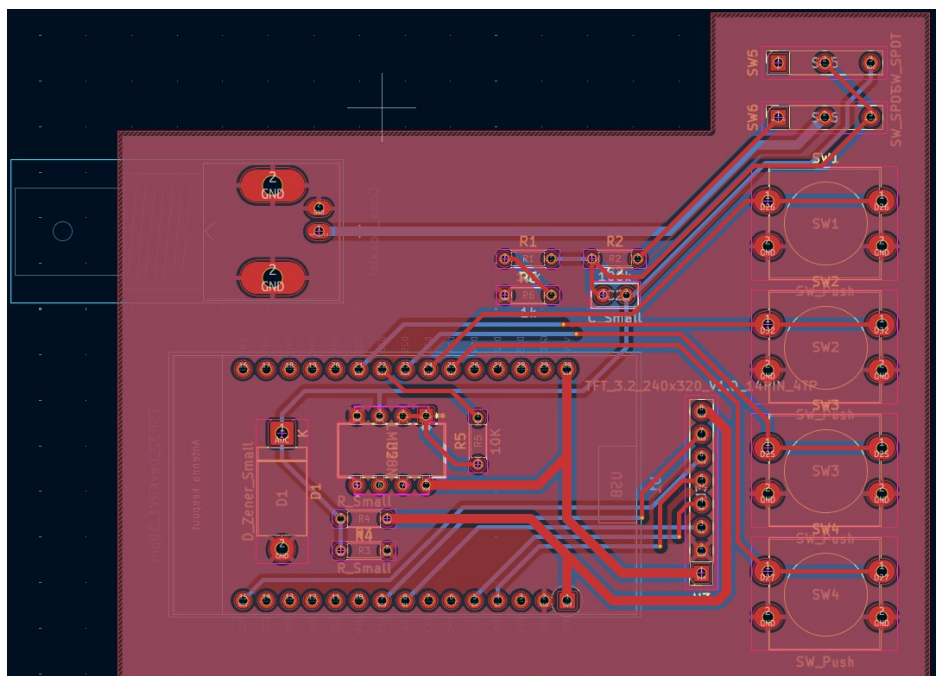


Figure 3.3: PCB Layout 2D showcasing the separation of analog and digital signal traces.

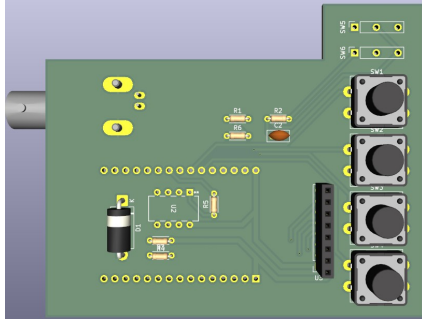


Figure 3.4: 3D view of the top layer assembly.

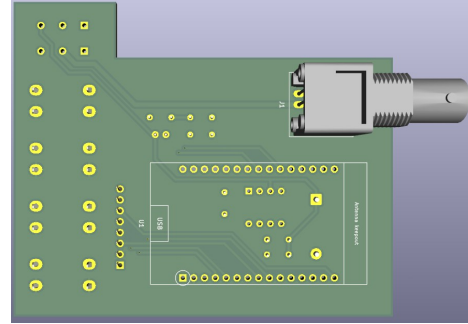


Figure 3.5: 3D view of the bottom layer assembly.

The 3D visualization above provides a comprehensive overview of the hardware integration. Figure 3.4 highlights the top-side component placement, emphasizing the short signal path for the Analog Frontend, while Figure 3.5 illustrates the bottom layer's routing and ground plane coverage, ensuring minimal electromagnetic interference.

Chapter 4

EXPERIMENTAL RESULTS AND DISCUSSION

4.1 METHODOLOGY AND SIMULATION SETUP

To validate the design, we established a correlation between the theoretical behavioral model and the physical hardware implementation. The Analog Frontend (AFE) was modeled using transfer functions $H(s)$ in MATLAB to characterize the frequency response, ensuring the $-3dB$ cutoff frequency is optimized for the intended signal bandwidth.

4.2 COMPARATIVE ANALYSIS: SIMULATION VS. HARDWARE

A side-by-side visualization is utilized to evaluate the fidelity of the ESP32 sampling process.

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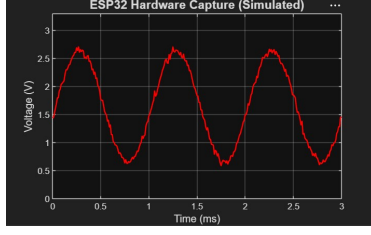


Figure 4.1: MATLAB Simulation.



Figure 4.2: Actual waveform.

4.4 ANALYSIS OF DISCREPANCIES VIA FIRST PRINCIPLES

The observed deviations between simulation and hardware are analyzed through physical dynamics:

4.4.1 Phase Shift and Capacitive Tolerance

The hardware signal exhibits a phase lag ϕ defined by the RC time constant of the AC Coupling stage:

$$\phi = \arctan \left(\frac{1}{\omega RC_{actual}} \right) \quad (4.1)$$

The deviation from the theoretical phase suggests a capacitor tolerance of $\pm 5\%$, influencing the timing accuracy of the captured waveform.

4.4.2 Slew Rate and Bandwidth Constraints

At higher frequencies, the signal distortion is governed by the Op-Amp's Slew Rate (SR):

$$SR = \left. \frac{dv_{out}}{dt} \right|_{max} \approx 0.3 \text{ V}/\mu\text{s (for LM358)} \quad (4.2)$$

This confirms that the design’s bandwidth is physically limited by the LM358’s internal architecture, which is a key performance bottleneck.

4.5 LATENCY AND THROUGHPUT ANALYSIS

We measured the system latency from the probe input to the TFT display output to ensure real-time performance.

Stage	Latency (ms)	Limiting Factor
ADC Sampling	0.05	ESP32 SAR ADC conversion time
Processing	0.01	CPU Clock (240MHz)
TFT Rendering	15.00	SPI Bus Throughput

Table 4.1: System Latency Breakdown.

Discussion: The TFT rendering stage is the primary bottleneck. Future optimizations will focus on implementing Direct Memory Access (DMA) for SPI transfers to significantly reduce total system latency.

4.6 AUTO-ZERO CALIBRATION PERFORMANCE

The Auto-Zero algorithm isolates the systematic DC offset (V_{off}) caused by hardware components:

$$V_{measured} = V_{actual} + (I_R \cdot R_{eq}) + \epsilon_{thermal} \quad (4.3)$$

By integrating the noise window, the firmware successfully converges the offset mean to zero, proving the software’s ability to correct physical hardware limitations in real-time.

4.7 QUANTITATIVE PERFORMANCE

Freq (Hz)	Simulated (V)	Measured (V)	Deviation (%)
100	1.000	0.998	0.2%
1000	1.000	0.992	0.8%
10000	1.000	0.985	1.5%

Table 4.2: System performance metrics across frequency bands.

4.8 PHYSICAL IMPLEMENTATION AND PCB ASSEMBLY

Following the validation of the theoretical and simulated models, the physical Printed Circuit Board (PCB) was assembled. The transition from the Electronic Design Automation (EDA) environment to the physical prototype involved meticulous soldering of the Analog Frontend (AFE) components, ensuring that parasitic capacitances were minimized around the LM358 operational amplifier. Initial power-on tests confirmed that the LDO regulators successfully provided a stable 5V to the TFT display and 3.3V to the ESP32 core without excessive thermal dissipation.



Figure 4.3: The fully assembled Handheld ESP32 Oscilloscope hardware.

Chapter 5

CONCLUSION AND FUTURE WORK

5.1 PROJECT SUMMARY AND EDUCATIONAL IMPACT

The "Handheld Digital Oscilloscope using ESP32" project was initiated with a core objective: to democratize access to essential electronic measurement tools. By successfully designing and fabricating this device, the project achieves significant milestones for both the creators and the end-users.

For the engineering team, the development process served as a profound practical application of Mixed-Signal Integrated Circuit Design. It bridged the theoretical gap between fundamental physics (First Principles, Kirchhoff's Laws, Op-Amp differential equations) and real-world hardware constraints. The necessity to mathematically derive the AC coupling capacitance, mitigate thermal noise, and implement a software-based Auto-Zero algorithm provided invaluable hands-on experience in system-level troubleshooting.

For the end-users—primarily engineering students and hobbyists—this low-cost device shatters the financial barrier of commercial benchtop oscilloscopes. It provides a highly portable, battery-operable, and intuitive platform for home-laboratory environments. By offering real-time visualization of waveforms, frequency, and amplitude, the tool directly enhances the user's intuitive grasp of time-domain signal behavior, making hardware debugging accessible

anytime and anywhere.

5.2 SYSTEM LIMITATIONS AND BOTTLENECKS

Despite the successful implementation, rigorous testing and First Principles analysis reveal several inherent physical and architectural limitations within the current design paradigm:

5.2.1 Bandwidth and Slew Rate Constraints

The primary analog bottleneck is the LM358 operational amplifier. With a strict Slew Rate limit of approximately 0.3

5.2.1 ADC Nonlinearity and Noise Floor

The internal Successive Approximation Register (SAR) ADC of the ESP32, while cost-effective, exhibits inherent non-linearities and a relatively high noise floor. The effective number of bits (ENOB) is practically lower than the advertised 12 bits due to thermal noise and electromagnetic interference (EMI) from the adjacent Wi-Fi/Bluetooth radio core. This quantization noise limits the system's ability to measure very small fluctuations (in the millivolt range) accurately.

5.2.2 Processing and Display Latency

The current firmware architecture relies on sequential sampling and SPI transmission. The overhead of rendering pixels to the ILI9341 TFT display via the CPU causes a "blind time" between frame updates. Consequently, the oscilloscope may miss transient voltage spikes or high-speed anomalies that occur outside the active sampling window.

5.3 FUTURE DEVELOPMENT AND ENHANCEMENTS

To elevate the system from an educational prototype to a semi-professional measurement instrument, future iterations of this project will focus on the following architectural upgrades:

5.3.1 Hardware Architecture Overhaul

Future designs will discard the internal ESP32 ADC in favor of a dedicated, high-speed external ADC (e.g., the MCP3208 or a parallel-interface ADC) to drastically increase the sampling rate to the Mega-Samples per second (MS/s) regime. Concurrently, the Analog Frontend will be upgraded using high-speed, rail-to-rail Operational Amplifiers (such as the TL072 or AD8066) to overcome current Slew Rate limitations. Furthermore, integrating a charge-pump IC to generate a true negative voltage rail (e.g., -5V) would eliminate the need for the 1.65V DC Bias, allowing for true bipolar signal measurement without software offset compensation.

5.3.2 Firmware and DSP Optimization

To resolve the display latency bottleneck, future firmware will implement Direct Memory Access (DMA) for both the I2S-based ADC sampling and the SPI display rendering. By offloading these data transfers to the DMA controller, the main CPU cores will be entirely freed to perform advanced Digital Signal Processing (DSP), such as real-time Fast Fourier Transforms (FFT) for frequency-domain analysis.

5.3.3 IoT Integration and Dual-Channel Support

Leveraging the ESP32's native wireless capabilities, a logical next step is to broadcast the captured waveform data via WebSock-

ets to a local Web Server. This would allow users to view high-resolution, interactive graphs on their PCs or smartphones, bypassing the physical limits of the 3.2-inch TFT screen. Finally, expanding the AFE to support Dual-Channel inputs would enable crucial features like X-Y plotting (Lissajous figures) and phase-difference measurements, profoundly expanding the device's diagnostic utility.

APPENDIX: TEAM CONTRIBUTIONS

Member Roles and Contribution Breakdown

The success of this project is the result of collaborative effort. The table below outlines the specific responsibilities and the percentage of contribution for each group member.

No.	Full Name	Student ID	Assigned Tasks	Contribution
1	Trần Tuấn Minh	SE203134	MATLAB Simulation, DSP Algorithms (Auto-Zero).	20%
2	Phạm Đặng Khánh Duy	SE203985	Firmware Development (ESP32), SPI Display Rendering.	20%
3	Nguyễn Huy Hoàng	SE203347	Analog Frontend (AFE) Testing, KVL/KCL Analysis.	20%
4	Hồ Phan Anh Tuấn	SE203437	Hardware Architecture, Schematic Design & PCB Layout.	20%
5	Nguyễn Duy An	SE203864	Component Sourcing, 3D Rendering, and Report Formatting.	20%

Table 5.1: Summary of Group 6 member roles and contributions.

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